Rony Zeenaldeen

Senior Backend & AI Architect

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SUMMARY

High-impact Senior Backend & AI Architect (4+ years) building production-grade, event-driven systems and applied AI at scale. Polyglot expert in Django, Go, and FastAPI, orchestrating Kafka-driven pipelines and Kubernetes infrastructure for high-throughput, fully observable platforms. Proven impact: 40× faster queries, 60–70% API speedup via Redis caching, and 65% reduction in database load.

TECHNICAL SKILLS

Back-End: Django, Django REST Framework, FastAPI, Go (Golang), Node.js, REST API Design, GraphQL, WebSockets

AI / ML: Python, TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, EfficientNetV2, Facebook Prophet, Transformer Models

Databases: PostgreSQL, Redis, MongoDB, MySQL, Firebase Firestore

Messaging & Streaming: RabbitMQ, Apache Kafka

Cloud & AWS: IAM, S3, RDS, SES, Route 53, Elastic Beanstalk

DevOps & Infrastructure: Docker, Kubernetes (Helm, k3s, Rancher-ready), NGINX, GitHub Actions, CI/CD, Linux

Monitoring & Reliability: Prometheus, Grafana, pgBackRest, pgBadger

Mobile: Flutter, Dart, GetX, BLoC, Firebase

Programming Languages: Python, Go, TypeScript, Dart, SQL, JavaScript

PROFESSIONAL EXPERIENCE

Senior Backend & AI Architect (Independent) Nov 2024 – Present

Architected and deployed production-grade AI systems, event-driven infrastructure, and secure enterprise automation tools for independent SaaS platforms and global clients.

- **Enterprise AI Document Automation (Client – Austria / Remote):** Engineered an end-to-end Django & Claude Vision VLM pipeline transforming scanned Arabic official documents into formatted DOCX. Collapsed a multi-stage OCR/LLM stack into a single VLM call returning strict JSON, eliminating error compounding, and overriding python-docx/Word XML limitations by directly manipulating raw <w:tc> trees and merging split text runs to preserve exact fonts and table layouts.
- **Intelligent Human-in-the-Loop Routing:** Designed an automated routing workflow using per-field VLM confidence scores (0.0–1.0), passing clean data instantly while dynamically routing uncertain fields (<0.85) to role-based human reviewers (Employee/Reviewer/Admin) with full audit logging.
- **Production AI & Event-Driven Infrastructure:** Built decoupled FastAPI inference microservices and managed Apache Kafka/RabbitMQ pipelines for heavy streaming and batch AI workloads; established full system observability using Prometheus and Grafana alerting with pgBackRest for point-in-time recovery.
- **High-Impact Applied ML Models:** Developed a chest X-ray classifier (EfficientNetV2S) achieving AUC-ROC 0.997 / F1 0.95 deployed via Django REST API, a sales forecasting pipeline (Facebook Prophet) on 1M+ transactions, and an HR attrition model yielding 87% accuracy and $225K–$450K in estimated annual retention savings.

Backend Systems Engineer (Independent) Jan 2022 – Nov 2024

Designed and delivered scalable, polyglot backend systems and event-driven architectures for production applications.

- **API Design:** Built and shipped secure REST APIs — via Django REST Framework and Go — with JWT/OAuth2 authentication, versioning, and OpenAPI documentation, plus GraphQL endpoints for flexible client queries.
- **Performance & Caching:** Reduced PostgreSQL query response times from 2000ms to <50ms, cut database load by 65%, and boosted API latency by 60–70% through strategic Redis caching.
- **Real-Time Streaming:** Designed real-time systems (WebSockets, Django Channels, Redis) for live ride-matching and location streaming.
- **Payments Integration:** Integrated Stripe, PayPal, and Razorpay with secure webhook handling and transaction processing.
- **Data Reliability:** Deployed Go microservices for latency-critical, high-concurrency components (WebSocket hubs, event consumers), with RabbitMQ/Kafka decoupling asynchronous processing and pgBackRest/pgBadger strengthening PostgreSQL backup and query analysis.
- **Cloud & DevOps:** Deployed backend services using Docker, Kubernetes, NGINX, and AWS infrastructure for high availability and scalability.

KEY PROJECTS

NeuroSeek AI — Agentic AI Research Platform **(LIVE)** · 2026 – Present [↗ Link]

Self-hosted research assistant that plans multi-step search queries, crawls and embeds web sources, and answers via hybrid RRF-fused RAG (pgvector semantic + keyword search) — streaming a fully cited report token-by-token.

- Implemented an agentic "gap-driven" re-search loop: the system audits its own answer coverage, identifies gaps, and autonomously re-searches before finalizing a report.
- Engineered a polyglot backend — Django 5 + DRF + Celery/Redis for the application layer, with dedicated Go services for a real-time WebSocket streaming hub and a Kafka consumer powering the analytics pipeline.
- Deployed cloud-native via Docker Compose and Kubernetes (Helm, k3s/Rancher-ready) with full observability — Apache Kafka (KRaft) event streaming, Prometheus/Grafana alerting, and pgBackRest (PITR) + pgBadger for database reliability — and hardened with SSRF protection, JWT rotation/blacklisting, and rate limiting.

Tech: Python · Django 5 · DRF · Celery · Redis · Go · PostgreSQL 16 · pgvector · Apache Kafka · Ollama · Next.js 14 · TypeScript · Docker · Kubernetes · Prometheus · Grafana · pgBackRest

PulmoSight — Doctor-in-the-Loop Chest X-Ray AI **(LIVE)** · 2026 – Present [↗ Link]

Self-hostable clinical AI system that classifies 18 chest pathologies with Grad-CAM heatmaps and drafts a structured radiology report grounded strictly in the classifier's confirmed findings, built for clinics that can't afford commercial radiology AI.

- Built a redaction-filter safety layer that strips any AI-generated finding the classifier didn't confirm — no report reaches a patient without a doctor's review and sign-off via an auditable approval workflow.
- Designed the detection pipeline: an ensemble of fine-tuned DenseNet-121 classifiers with per-disease F1-tuned thresholds and a grounded VLM report writer (switchable Qwen2-VL-2B / Phi-3.5-vision / MAIRA-2 backend).
- Architected a polyglot backend — Django 5 + DRF for the clinical app and audit logging, a separate FastAPI inference service, and a Go-based Telegram bot (RabbitMQ-decoupled from doctor approval) delivering a bilingual (Arabic RTL / English) report via QR + single-use PIN onboarding, consent capture, and a branded PDF.

Tech: Python · Django 5 · DRF · FastAPI · Go · RabbitMQ · PostgreSQL 16 · PyTorch · Grad-CAM · Docker · Telegram Bot API

ROTECHAI — AI + Software Studio Platform **(LIVE)** · 2026 – Present [↗ Link]

Self-hosted, multilingual (EN/AR/NL/DE) marketing site paired with a live AI product — real-time text classification and a RAG chat assistant for services, pricing, and process questions.

- Right-sized the entire AI layer to a single ~400MB multilingual embedding model (fastembed / ONNX Runtime, no PyTorch, lazy-loaded) with a hybrid cosine + keyword RAG over an in-memory knowledge base — no GPU or paid LLM API required.
- Designed the backend as Django 5 + DRF for the main application paired with an independently restartable FastAPI microservice for AI inference, so an AI-service restart never takes the site down.
- Built a Next.js 16 (App Router) + React 19 + TypeScript frontend across 4 locales with full RTL Arabic support and SEO best practices (structured data, hreflang, sitemaps).
- Hardened and deployed the full stack behind a single NGINX edge with HSTS, CSRF/cookie hardening, brute-force lockout, rate limiting, and environment-only, fail-closed secrets.

Tech: Python · Django 5 · DRF · FastAPI · PostgreSQL 16 · Redis · Next.js 16 · React 19 · TypeScript · Docker · NGINX

Enterprise AI Translation & Automation Platform (Confidential Client – Austria) · April 2026 – Present

An enterprise-grade system that transforms scanned Arabic official documents into perfectly formatted, translated DOCX files with integrated human-in-the-loop validation.

- Engineered a single-call VLM pipeline via Claude Vision to extract fields, classify documents, and output structured, translated JSON, eliminating a multi-hop OCR/LLM stack prone to error compounding.
- Overrode low-level python-docx limitations by parsing raw <w:tc> XML trees and merging split text runs to fix corrupted template rendering in merged tables, ensuring full layout and font fidelity.

Tech: Python · Django 5 · DRF · Claude Vision (Anthropic) · python-docx · pypdfium2 · PostgreSQL · JWT · Docker

EDUCATION

B.Sc. Informatics Engineering — Software & Applications Development

Al-Shahba Private University, Aleppo, Syria · Sep 2020 – Jun 2025

CERTIFICATIONS

- Artificial Intelligence Essentials V2 — IBM (2025) [Link]
- Data Engineering Professional — IBM (2025) [Link]
- Strategic Planning Professional Certificate — Canada [Link]
- Project Management Professional (PMP v6) — Project International (2021) [Link]

LANGUAGES

- Arabic — Native
- English — B2-C1 (Proficient)